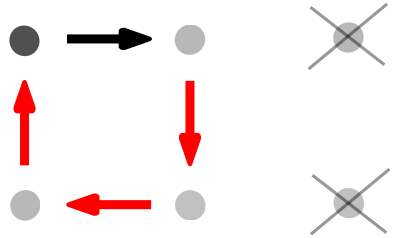
Q. What is the most concentration-sensitive promoter architecture?

sharp response?
cooperative?
proofreading?

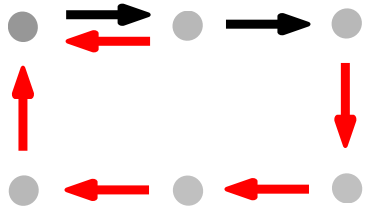
A. It depends on the limiting constraint

1. Cooperativity only optimal above critical concentration

"Keep binding sites free" (diffusion-limited)



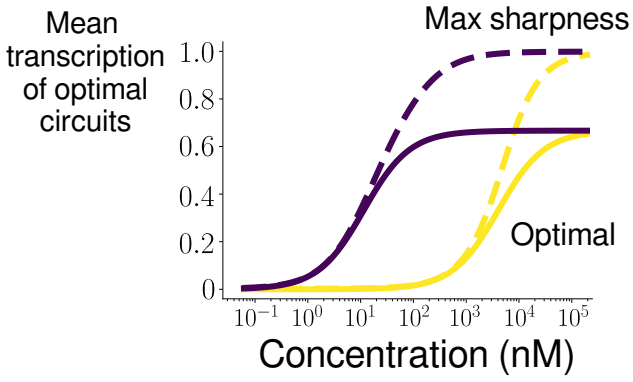
"Bind cooperatively" (activation-limited)

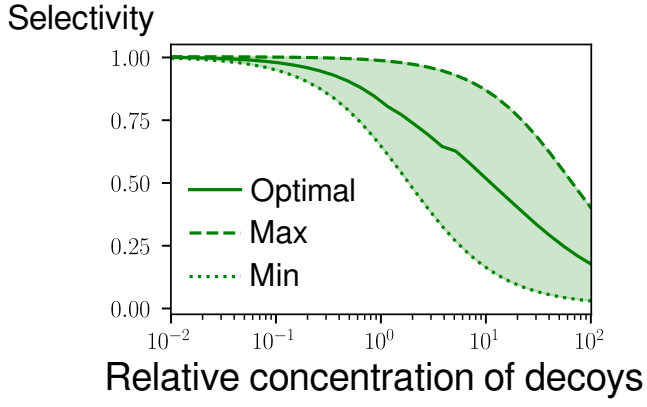


Proofreading only optimal above critical **decoy** concentration

2. Solutions sets are generically **discontinuous**

3. Optimal promoters neither maximize sharpness nor selectivity





Discussing optimality requires precise accounting of underlying constraints